

Xinbao Qiao

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Education

The Chinese University of Hong Kong Ph.D., Information Engineering. Advisor: [Angela Yingjun Zhang](#). *Hong Kong. 2026~present*

Research Areas My research primarily studies lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted.

Teaching AI Algorithms and Systems. Secure Artificial Intelligence.

Zhejiang University M.Sc. in Artificial Intelligence. Advised by [Meng Zhang](#).

Zhejiang. Sep 2022 ~ Dec 2025

Shandong University B.Eng. in Communication Engineering.

Shandong. Sep 2018 ~ Jul 2022

Experience

Doctoral Research @ The Chinese University of Hong Kong

Hong Kong. 2026 ~ present

- Study data-centric ML and AI-network co-design, including distributed Wasserstein barycenter methods for shared references without pooling raw data. **Paper #5**

Research on Data-Centric ML Systems @ Zhejiang University

Zhejiang. Mar 2023 ~ Dec 2025

- Developed soft-weighted unlearning methods that continuously correct data influence for fairness and robustness without binary erasure. **Paper #2**
- Designed Hessian-free online certified unlearning methods using recollected trajectory statistics instead of explicit Hessian inversion, supporting online deletion requests without second-order solves. **Paper #3**
- Studied exact and efficient random-forest unlearning in dynamic online environments, preserving predictive utility under incremental data updates. **Paper #4**

Research on Trustworthy LLM Systems @ NUSRI CQ

Chongqing. Jun 2025 ~ Dec 2025

- Analyzed low-resource sample selection in recursive synthetic-data training and proposed collaborative Wasserstein evaluation proxies. **Paper #1**
- Examined illusory pattern perception in LLMs and developed an interpretable framework for spurious inference behavior and misleading responses. **Paper #6**

Open-Source Contributions and Services

🔗 **Research code releases**, public code for certified unlearning, soft-weighted unlearning, and model-collapse work.
🏠 **Xinbaopedia**, public academic homepage and wiki-style research archive with papers, figures, CV, and project notes.
👤 **Peer-Reviewing**, reviewer for ICML 2026, NeurIPS 2025 and 2026, ICLR 2025, AAAI 2025 and 2026, and IEEE TNNLS.

Selected Publications

Asterisks (*) denote co-first authorship. Accepted papers are listed before under-review manuscripts.

1. When **Sample Selection Bias** Precipitates **Model Collapse**

Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang

Low-resource verification can turn local sample selection into persistent tail pruning.

📄🔗🏠 ICML 2026.

2. Beyond Binary Erasure: **Soft-Weighted Unlearning** for Fairness and Robustness

Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang

Soft-weighted corrective unlearning for non-binary fairness and robustness interventions.

📄🔗🏠 AAAI 2026.

3. **Hessian-Free Online Certified Unlearning**

Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei

Certified unlearning without explicit Hessian inversion through recollected statistics.

📄🔗🏠 ICLR 2025.

4. **DynFrs: An Efficient Framework for Machine Unlearning in Random Forest**

Shurong Wang, Zhuoyang Shen, Xinbao Qiao, Tongning Zhang, Meng Zhang

Exact and efficient random-forest unlearning in dynamic environments.

📄🔗🏠 ICLR 2025.

5. **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification**

Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang

Barycentric multi-prototype classification for federated representation geometry.

🏠 Under review.

6. **Illusory Pattern Perception Drives Spurious Inference in Large Language Models**

Peihua Mai, Zhuoyan Shao, Xinbao Qiao, Meng Zhang, Xinyue Zhou, Yan Pang

Analysis of spurious pattern perception and interpretable behavior in LLMs.

🏠 Under review.